- Libraries offer a wide range of modules and functions that support fast application development.
- Easily integrated with other languages like Java, C/C++, etc.
- Provides interfaces for a large set of databases.
- Supports object-oriented style of programming that encapsulates code within the object.
- Has a variety of basic data types like integer, floating point number, string (both in ASCII and Unicode), list, dictionaries, etc.
- Code can be grouped into packages and modules.
- Automatic memory management deallocates the memory instead of manually handling it in the code.
- Web applications can be developed using Python. It provides libraries to handle protocols like HTML, XML, JSON, requests, etc. Various frameworks such as Django, Pyramid, etc, are available for web application development.
- Provides various libraries and packages like SciPy, Pandas, IPython, etc, for developing numeric and scientific computations using Python.

Python libraries for AI/ML

The most popular Python libraries for AI/ML development are:

- **Scikit-learn** for data mining and analysis, which optimises Python's machine learning usability.
- **NumPy,** which is used for scientific calculations.
- **SciPy,** which is used for advanced computation covering natural language processing (tokenization, named entity recognition, parsing).
- **Pandas** offers developers high-performance structures and data.
- **TensorFlow** for building an end-to-end deep learning platform for production tools and the deployment ecosystem.
- **PyTorch,** a deep learning library with strong production tooling used to build deep learning models.
- **MLflow,** which is used for model registry and deployment tooling for ML lifecycle management.
- **LightGBM,** a memory-efficient gradient-boosting library from Microsoft with GPU support.
- **Keras,** a high-level deep learning API used in model prototyping and training.

Best practices for using Python

Many seasoned Python developers find themselves coding on autopilot—relying on patterns learnt years ago simply

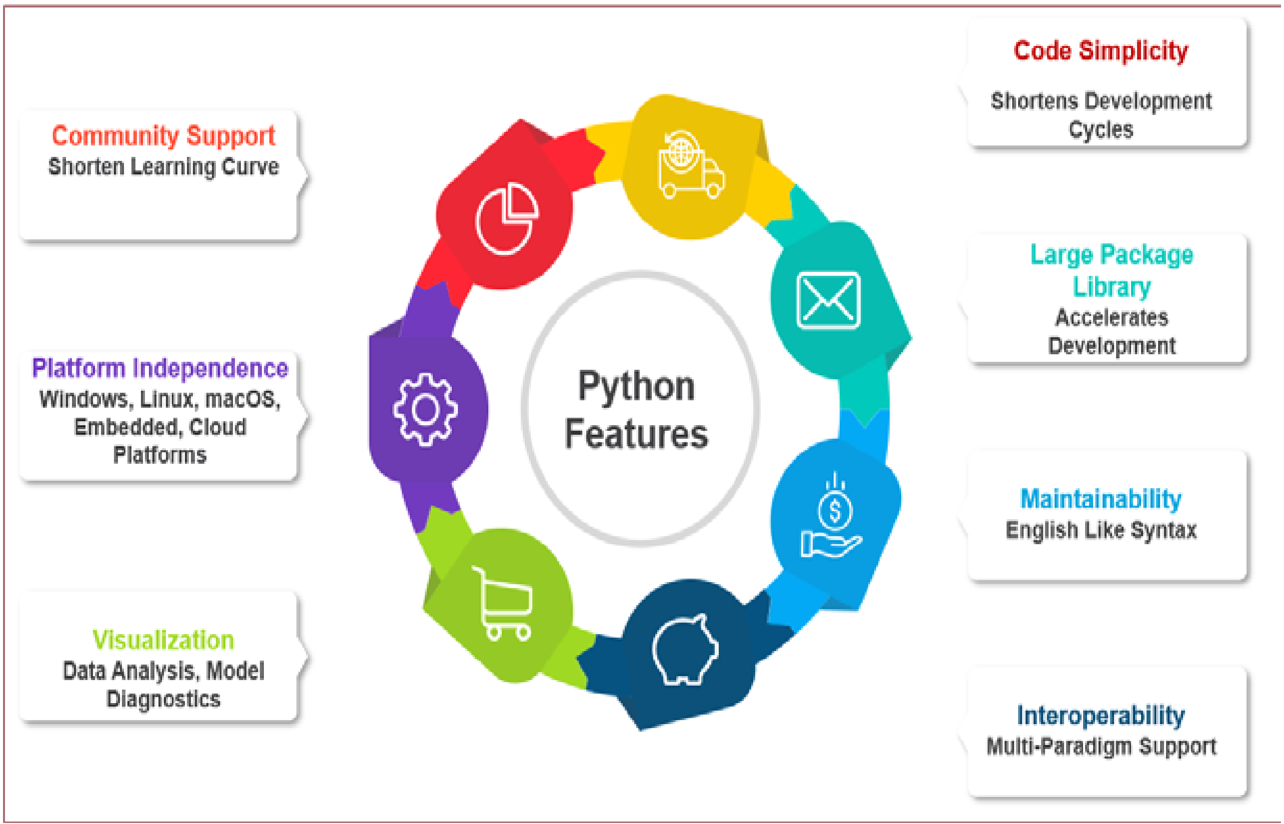

Figure 1: Features of Python